NGIAB-NRDS Demo

NextGen In A Box –NRDS Ecosystem

Alabama Water Institute
CIROH · Lynker · BYU · Aquaveo · USU · RTI

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4 Years of Progress

NGIAB-NRDS Demo

- Ngen OWP Forks
- NGIAB Evolution
- NRDS
- Hub
- R2OHC Platform
- CCNH
- What's Next

April 2026

THE UNIVERSITY OF ALABAMA | Alabama Water Institute

CIROH Cooperative Institute for Research to Operations in Hydrology

Today's Agenda

A structured look at four years of open-source hydrology work

01 **Ngen Forks**
CIROH-UA forks of ngen and t-route powering NGIAB

02 **NGIAB Evolution**
From v1 to today — architecture, containers, CI/CD

03 **NGIAB Demo**
Live walkthrough of the current NGIAB stack

04 **CCNH Project**
Cloud-based JupyterHub platform

05 **Cyberinfrastructure**
TEEHR, Tethys, R2OHC — the platform layer

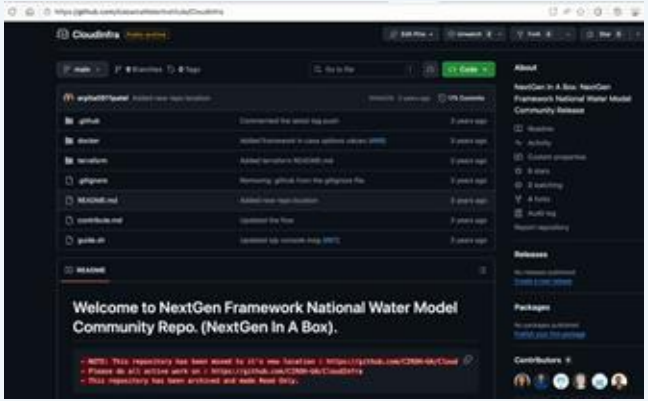
06 **NRDS Project**
NextGen research datastream - daily continental runs

07 **What's Next**
Roadmap & open questions

08 **Your Feedback**
Discussion & input from Ngen devs and NOAA leadership

Four Years at a Glance

From first container to continental-scale daily runs



2023

NGIAB First Docker container in AWI GitHub
4 contributors

Early 2023

GitHub Actions
ARM64 + AMD64

2024

Data Preprocess, DataStreamcli

R2OHC Platform

Tethys Visualizer integration

2025

NRDS prototype

TEEHR Integration

2026

CIROH **Hub**

CCNH project

NRDS Scaling

14K+ downloads

14,400+	1180+	60+	400+	3
Docker pulls (NGIAB)	Daily NRDS executions	Community repositories	Active Users of R2OHC Platform	Deploy targets (local/cloud/HPC)

CIROH-UA GitHub

CIROH-UA GitHub Enterprise Organization

CIROH-UA

Overview | Repositories 74 | Discussions | Projects 12 | Packages | Teams 2 | People 12 | Security and quality | Insights | Settings

CIROH-UA

CIROH-UA enterprise github account - Advancing water resources modeling through open-source innovation

62 followers · United States of America · https://ciroh.ua.edu/ · https://hub.ciroh.org · https://awi.ua.edu/

Part of The University of Alabama

README.md

CIROH-UA

Cooperative Institute for Research to Operations in Hydrology

The University of Alabama - Alabama Water Institute

Followers 62 | Website ciroh.org | Hub hub.ciroh.org

Mission

Advancing hydrology through open-source tools, community collaboration, and cutting-edge research.

Featured Projects

NextGen In A Box (NGIAB) Ecosystem

Containerized solution for running the NextGen Water Resources Modeling Framework locally

- NGIAB-Cloudless - Docker distribution for wide compatibility
- NGIAB-HPCinfra - Singularity distribution for HPC-optimized modeling
- NGIAB Data Preprocess - Data preparation tools
- NGIAB-TEEHR - NGIAB TEEHR integration
- NGIAB-Client - NGIAB Visualizer based on Tethys platform
- NGIAB Calibration - Calibration tools
- NGIAB 101 Training - A comprehensive guide to getting started with NGIAB Ecosystem
- Documentation on CIROH Hub - In-depth information on NGIAB and related tools
- Community Hydrofabric Patcher - Source for patched hydrofabric files retrieved by Data Preprocess
- PyNGIAB - Community NextGen Hub on 2i2c JupyterHub
- NGIAB Website - NGIAB product portfolio website - ngiab.ciroh.org

NextGen Research DataStream (NRDS)

Automated system for continuous hydrologic predictions

- forcingprocessor - Forcingprocessor converts National Water Model (NWM) forcing data into Next Generation National Water Model (NextGen) forcing data
- ngen-datastream - Infrastructure and tooling for the NextGen Research DataStream (NRDS) - Datastream Orchestration on AWS
- datastreamcli - DataStreamCLI is a stand alone tool that automates the complete workflow from preprocessing input data for NextGen to execution of the NextGen simulation through NextGen in a Box (NGIAB)
- S3 bucket - datastream.ciroh.org - README.html provides more details
- NRDS Visualizer - nrds.ciroh.org
- Community parameter integration capability (in-progress)

NextGen Framework Core Components

Essential modeling engines powering the NextGen Water Resources Modeling Framework

- ngen - NextGen Water Resources Modeling Framework Engine *(fork from NOAA-OWP repo)* (ngiab branch integrated into NGIAB)
- t-route - Tree-based hydrologic routing engine for river network flow *(fork from NOAA-OWP repo)* (ngiab branch integrated into NGIAB)
- lstm - Basic Model Interface (BMI) for streamflow prediction using Long Short-Term Memory (LSTM) networks (main branch - integrated into NGIAB)
- rust-lstm - A Rust implementation of a BMI adapter for LSTM-based streamflow prediction

Data Access

- api-nwm-gcp - NWM BigQuery API - REST API backed by National Water Model data, developed and deployed on Google Cloud Platform
- nwmurl - It provides utility functions specifically designed to subset and generate National Water Model (NWM) data URLs
- nwm_kerchunk - It provides National Water Model (NWM) output, which is stored as .nc files, into JSON headers and upload them to an AWS S3 bucket.

CIROH-2i2c JupyterHub

- awi-ciroh-image - CIROH JupyterHub image repository

https://github.com/CIROH-UA

NGIAB Evolution - Community Water Modeling

Open-source, containerized architecture, CI/CD, and extensions over 4 years

v1.1
2023

- Single-stage **Docker** build
- Manual dependency config
- **Local** machine only
- Basic **ngen + t-route with CFE+NOM**

v1.3
2024

- **Forked** NOAA-OWP ngen and t-route
- ngen-**datapreprocess** v1.0.0
- **Singularity** / HPC support
- **Multi-stage Docker** build

v1.7
2025

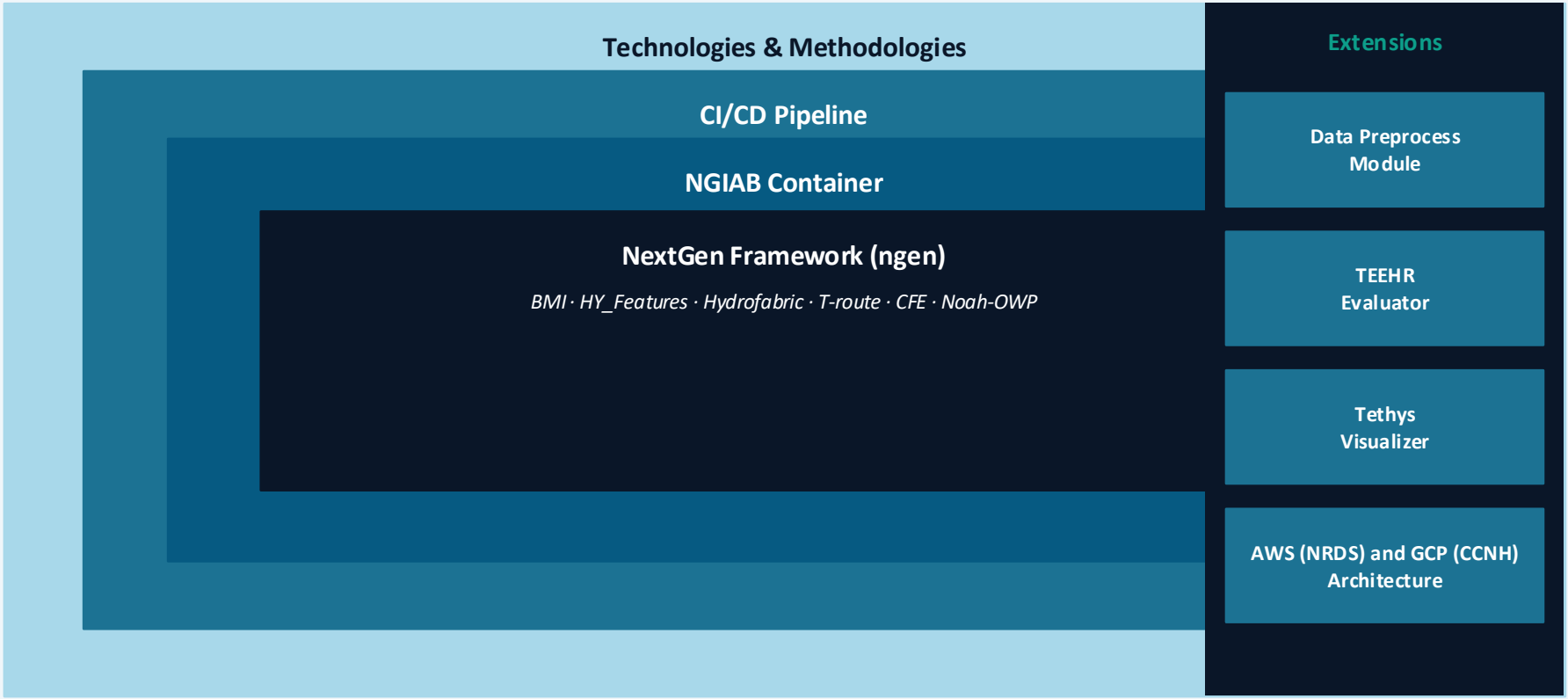
- NGIAB-**Calibration**
- NRDS prototype on **AWS**
- **LSTM** integration
- ngen-datapreprocess **GUI** v3.2.3

Now-v1.8
2026+

- **dHBV and SUMMA** integration
- TEEHR-Cloud (Apache Iceberg) - CONUS benchmarking on **AWS**
- Tethys web visualizer
- CCNH on 2i2c JupyterHub on **GCP**

NextGen-NGIAB-NRDS-CCNH Architecture

Layered design: framework → container → CI/CD → community



NGIAB Demo (DataPreprocess (GUI,CLI) + Guide script)

End-to-end: data prep → container run → evaluation → visualization

1 Pull Data

Select catchment + date range
Data Preprocess UI
S3 bucket download

2 Run NGIAB

Docker / Singularity
guide.sh for config
Serial or MPI parallel

3 Evaluate

TEEHR framework
NSE, KGE, bias metrics
vs USGS observations

4 Visualize

Tethys web app
Geospatial maps
Time series plots

📍 NGIAB 101 training module : https://ngiab.ciroh.org/training-NGIAB-101/

github.com/CIROH-UA/NGIAB_data_preprocess · github.com/CIROH-UA/NGIAB-CloudInfra · github.com/CIROH-UA/NGIAB-HPCInfra

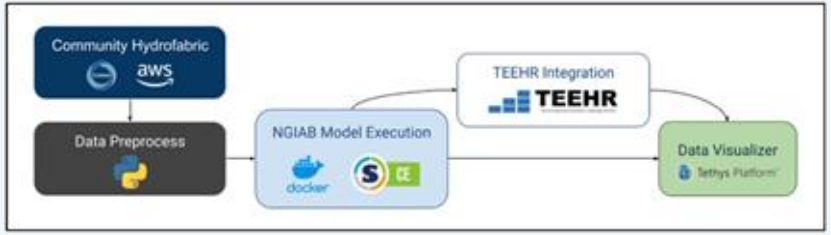
NGIAB Demo

End-to-end: data prep → container run → evaluation → visualization

NextGen In A Box (NGIAB)

Available Models

- Default Configuration: Conceptual Functional Equivalent (CFE, rainfall-runoff model)+ Noah-OWP-Modular (NOM, land surface model)+ t-route (Muskingum-Cunge routing model) (NOAA-OWP)
- Other NOAA-OWP models:
  - Long Short-Term Memory (LSTM, deep learning model with UA-contributed weights)
  - Soil Freeze-Thaw (SFT, soil heat transfer model)
  - Soil Moisture Profiles (SMP)
  - Potential Evapotranspiration (PET)
  - Lumped Arid/Semi-arid Model (LASAM, infiltration and rainfall-runoff model)
  - TOPMODEL (physical watershed model)
  - t-route diffusive wave routing model
- Community models:
  - Differentiable Hydrologiska Byråns Vattenbalansavdelning 2.0 (δHBV2, Penn State, differentiable rainfall-runoff model)
  - Structure for Unifying Multiple Modeling Alternatives (SUMMA, University of Calgary, hydrologic modeling framework)



1. Physical models
2. ML models
3. Hybrid models

12+ models integrated

Reduced Setup time:
Weeks to ~30 mins (97%)

Show NGIAB demo and Recordings on NGIAB website

NGIAB Demo

End-to-end: data prep → container run → evaluation → visualization

GUI:

uvx --from ngiab-data-preprocess **map_app**

./guide.sh

OR

CLI:

uvx --from ngiab-data-preprocess cli -i **cat-485385** --subset --start 2020-01-01 --end 2020-01-11 --forcings --realization --source aorc --lstm_rust --run

./runTeehrs script and ./viewonTethys script

More info: https://hub.ciroh.org/docs/products/ngiab/intro/at-a-glance/

CCNH — CIROH Community NextGen Hub on 2i2c-JupyterHub

A Cloud-based JupyterHub platform for community research with the NextGen water resources modeling framework

CCNH lowers technical and infrastructure barriers to participation in NextGen modeling by providing a cloud-hosted, end-to-end easy to use modeling research environment.

1 No local installation - Zero setup time

Pre-configured with NextGen, its dependencies, and a robust and extensible (CIROH produced) suite of compatible tools for NextGen model formulation, setup, calibration, evaluation (including TEEHR) and analysis.

2 Collaborative Open Research

Community-contributed model formulations (e.g. LSTM, custom BMI modules) run and validated against NGIAB baseline (CFE+Noah-OWP). Model runs and results are shared as reproducible, citable HydroShare resources, enabling reuse, transparency, and community collaboration.

3 Regional Case Studies

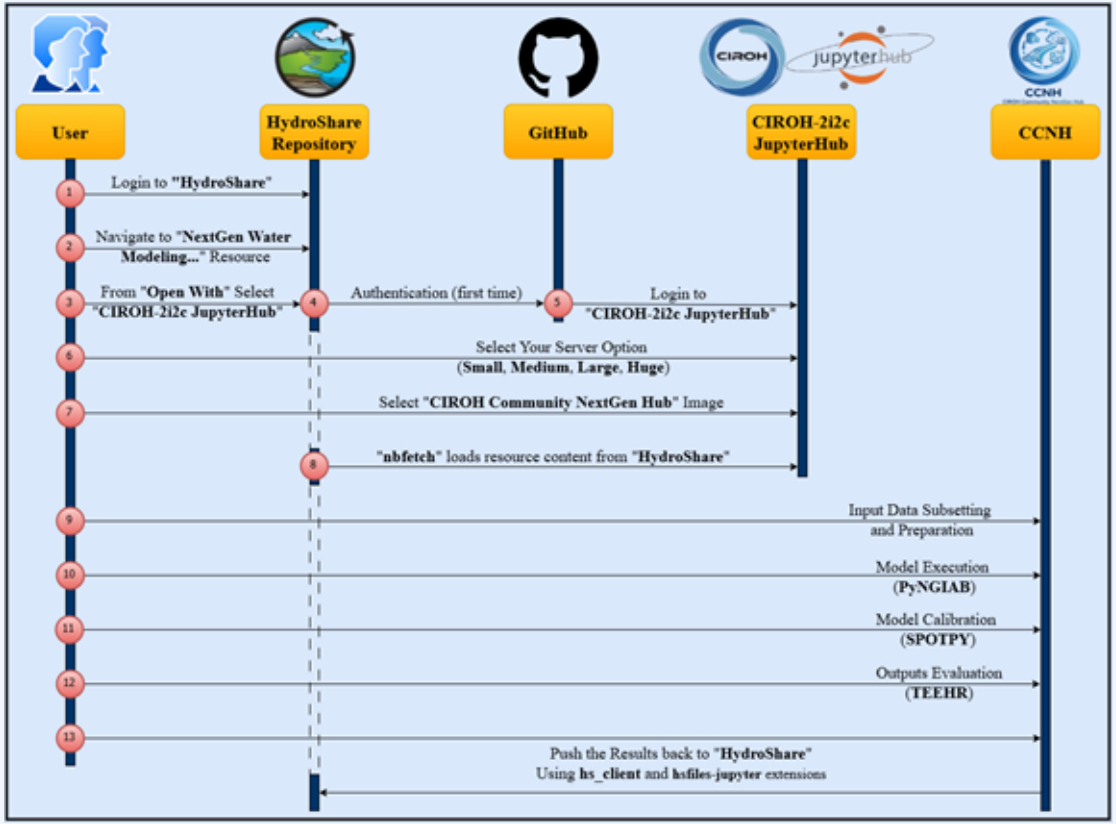
Researchers define spatial domains and simulation periods, generate model inputs, run NextGen formulations, and publish results as citable HydroShare resources that directly support and accompany peer-reviewed scientific publications.

4 Training & Outreach

Step-by-step Jupyter notebooks covering the complete NextGen modeling workflow — from watershed delineation through evaluation with TEEHR. Training materials hosted as resources on HydroShare. Accessible to any researcher with a HydroShare and CIROH authorized GitHub account.

CCNH — CIROH Community NextGen Hub on 2i2c-JupyterHub

Workflow for users to access, configure and run NextGen hydrologic simulations in a cloud-based environment



Documentation: https://hub.ciroh.org/docs/products/ngiab/distributions/nextgen-2i2c/

Live CCNH demo using 2i2c-JupyterHub

NextGen Research DataStream (NRDS)

Daily continental-scale NextGen runs — open, community-editable

NRDS By the Numbers

1180+
Daily executions

AWS
Cloud hosted

Open
Config & outputs

24/7
Continuous runs

What is NRDS?

An array of daily NextGen DataStream executions in the AWS cloud. Automates input collection, orchestrates NGIAB runs, and manages outputs. Configurations are open-source and community-editable — enabling regional experts to improve forecasts over their domains. Designed to accelerate the transition of novel modeling approaches from research to operational readiness.

Input Automation

Forcings, hydrofabric subsets, realizations auto-collected and formatted

NGIAB Orchestration

Spins up NGIAB containers, runs NextGen at scale, captures output

Open Outputs

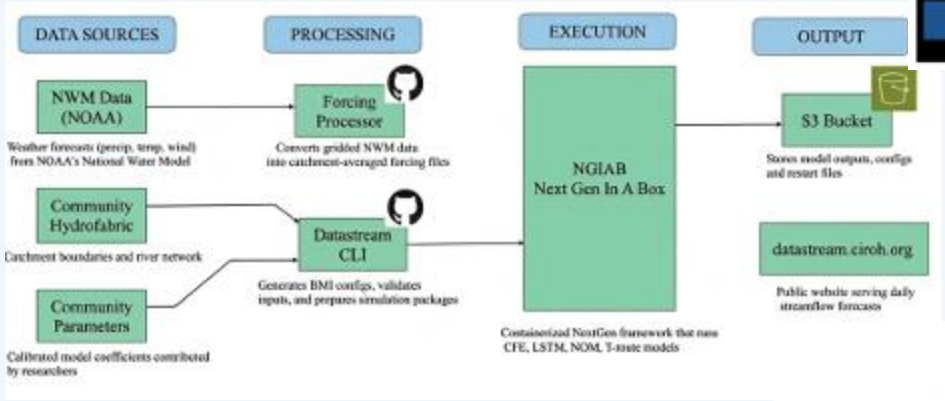
Forcings, configs, results published publicly for reproducibility

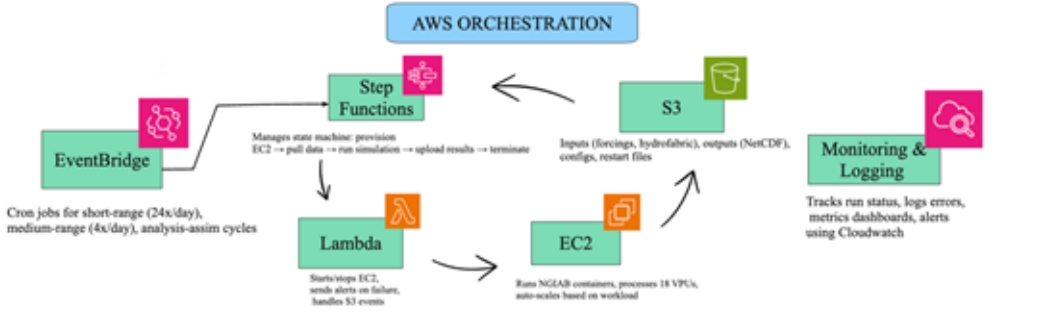
Community Config

Anyone can submit a regional config — improving local accuracy over time

NextGen Research DataStream (NRDS)

Daily continental-scale NextGen runs — open, community-editable





datastreamcli demo

For reproducing NRDS simulations Local/Cloud (done) HPC (in-progress):

```
$ cd datastreamcli

$ ./scripts/datastream -s 202006200100 -e 202006210000 -C NWM_RETRO_V3 -d
$(pwd)/data/demo -g https://ciroh\-community\-ngen\-
datastream\.s3\.amazonaws\.com/resources/v2\.1\_hydrofabric/geopackages/test\_data/pali
sade\.gpkg -R $(pwd)/configs/ngen/realization_sloth_nom_cfe_pet.json -n 4
```

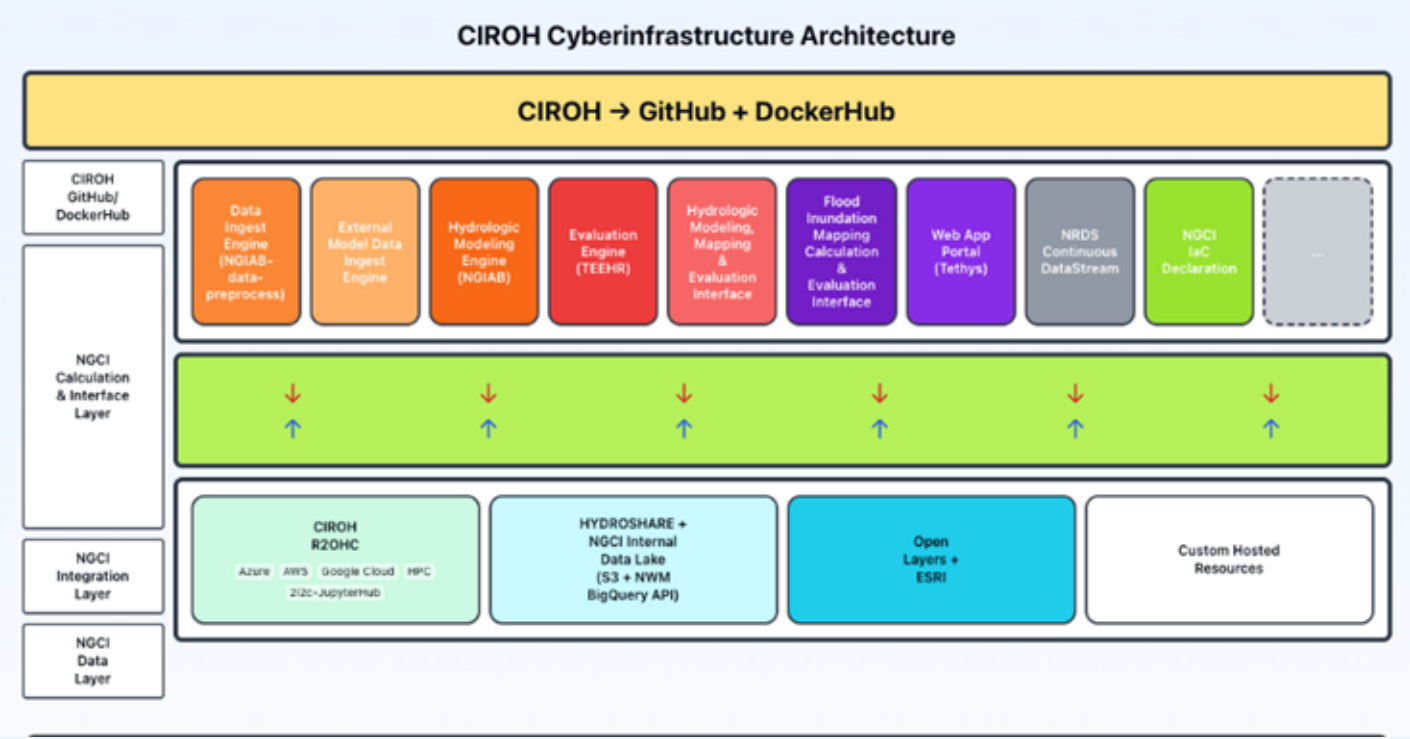
Options: https://github.com/CIROH-UA/datastreamcli/blob/main/docs/DATASTREAM_OPTIONS.md

Cyberinfrastructure – R2O Hybrid Cloud (R2OHC) Platform

Cloud Computing, HPC and local VMs

NGCI (NextGen Cyberinfrastructure)

The platform layer enabling big data, models, evaluation and visualization



R2OHC — Services to the Consortium

CIROH Research to Operations Hybrid Cloud — active support across 100+ projects & 500+ users

AWS

24 projects · 60+ users

EC2 · S3 · Lambda · ECS

- NRDS daily NextGen continental runs
- TEEHR application, Tethys Platform
- Open S3 buckets for forcing data & model outputs
- NGIAB CI/CD build infrastructure (ARM64 + AMD64)

GCP & 2i2c JupyterHub

63 projects · 180+ users

GCP · JupyterHub · Colab

- NWM API access & scalable batch compute
- 2i2c managed JupyterHub for collaborative research
- Interactive notebook environments for TEEHR evaluation
- Shared compute for consortium data science workflows

On-Premise HPC

20 projects · 75+ users

Pantarhei · Wukong · Slurm/PBS

- Singularity/Apptainer container execution on clusters
- Large-scale parallel NextGen simulations via MPI
- UA-hosted Pantarhei & Wukong cluster access
- OpenStack provisioning for reproducible environments

NSF ACCESS

7 projects · 108+ users

JetStream2 · Anvil · Bridges-2

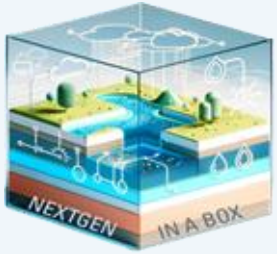
- National HPC allocations for CIROH consortium PIs
- JetStream2 cloud VMs for scalable model experiments
- Anvil & Bridges-2 for computationally intensive runs
- Portable NGIAB containers across ACCESS resources

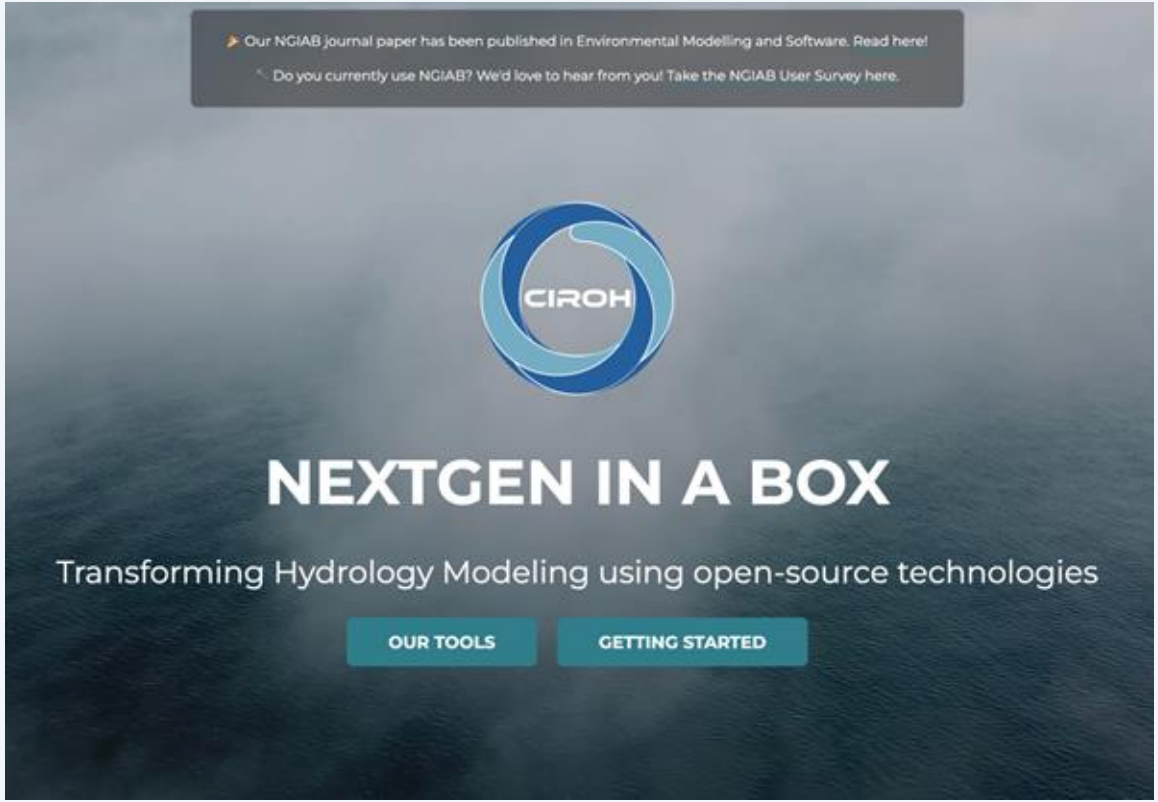
100+ active projects | **500+ total users** | **4 computing platforms - 270+ TB storage** | Portable · Reproducible · Open

NGIAB Website

One stop shop for everything related to NGIAB-NRDS ecosystem

ngiab.ciroh.org

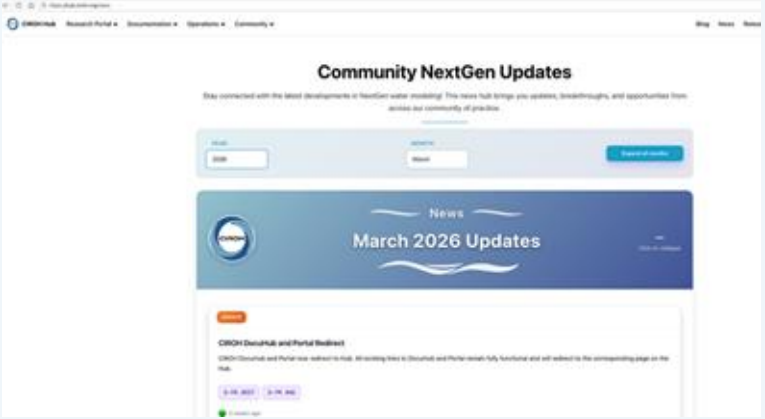


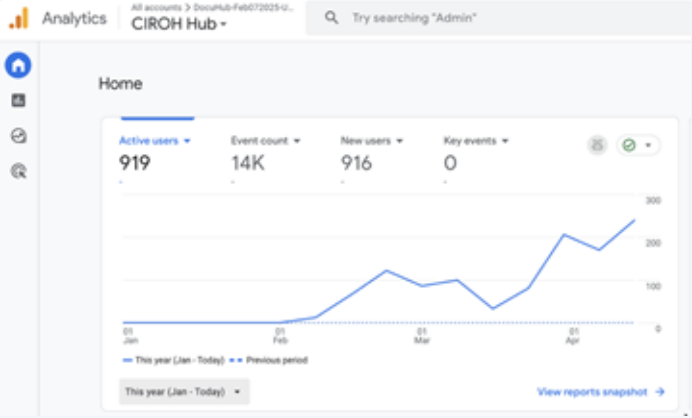


Website demo and NRDS dashboards

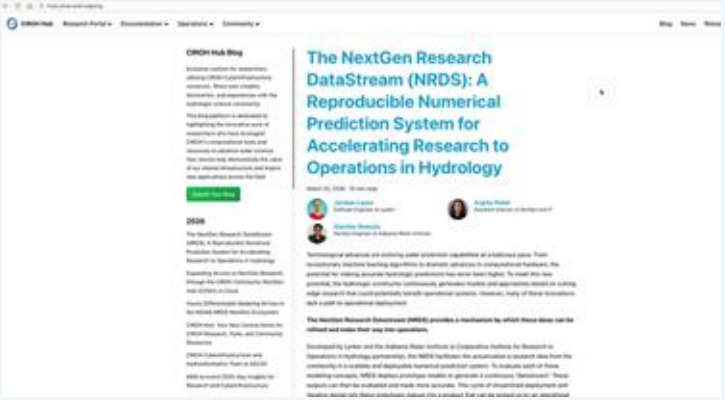
Hub – Community NextGen Blog, News and office hours

One stop shop for everything CIROH Cyberinfrastructure and Research tools





Browse CIROH Hub



What's Next

Near-term roadmap — phased releases across research teams

Q2 2026 *[In Progress]*

Visualization & UX

- → Enhanced Tethys visualizer for output
- → Improved catchment selection UI for NGIAB
- → Update onboarding documentation

Q2 2026 *[In Progress]*

Calibration & Performance

- → Automated calibration workflows in NGIAB
- → Expand TEEHR-NRDS integration and features

Q3 2026 *[Planned]*

NRDS Expansion

- → Increased domain coverage
- → Community contribution submission forms
- → Versioned output archive

2026+ *[Roadmap]*

NWM v4 related support

- → Align NGIAB with NWM v4 components
- → Official R2O pathway
- → Long-term CIROH support plan

Q&A

GitHub: *github.com/CIROH-UA*

Hub: *hub.ciroh.org*

NGIAB-NRDS: *ngiab.ciroh.org*

NGIAB 101: *https://ngiab.ciroh.org/training-NGIAB-101/*

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